Discrete Mathematics

Class: MATH 2410

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Standard Binary Encodings

Integers

Sign bit: The first bit is used to indicate the sign of the number, with 0 representing positive and 1 representing negative.
Two's complement: Negation is done by inverting all bits and adding 1.
Offset binary: Labeling the values of a binary number system from the minimum value to the maximum value, with the midpoint being zero. For example, in an 8-bit offset binary system, the values range from -128 to 127, with 0 represented as 10000000.

Real Numbers

Suppose 32 bits are used to represent a real number. The first bit is the sign bit, the next 8 bits are the exponent, and the last 23 bits are the mantissa. The exponent is stored in **offset binary** notation. The mantissa is stored in normalized form, meaning that the leading 1 is not stored.

General Tips

• $\sqrt{x^2} = |x|, x \in \mathbb{R}$ • $(\sqrt{x})^2 = x, x \geq 0$ • $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ • $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$
• $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$ • $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$ • $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$ • $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$